

HTML File User Guide

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Purpose of This Document

The HTML file provided with this document allows users to visualise the results obtained with the BaM (Bayesian Modelling) tool, and to compare them with the data supplied by hydrometric station managers. Most of the charts are interactive; a dedicated section therefore explains the available tools. In addition, 14 categories are available in each file and will be described below.

Interacting with the Charts

Displaying Legends

Chart legends are also interactive: they can be enabled or disabled by clicking on them. If the user wishes to display only one legend item — while all are currently enabled — they can do so by double-clicking on the desired item. Conversely, if several legend items have been disabled, they can all be re-enabled by double-clicking on any one of them. If the full legend is not visible, it can be scrolled using the cursor on the right side of the legend, or by scrolling with the mouse wheel.

Using the Interface Tools

Ten icons are available in the upper right corner of each chart for interacting with it.

Download plot as PNG

This tool (Figure 1) saves the chart as a PNG file, including any modifications previously made.



Figure 1: Saving plot as a PNG file

Zoom

This icon (Figure 2) enables manual zooming on the chart. It is particularly useful when another icon has been previously activated.



Figure 2: Zoom within the plot

Pan

This tool (Figure 3) allows the user to navigate within the chart, especially after zooming in.



Figure 3: Movement within the plot

It is also possible to navigate using the axis: when the cursor is moved close to an axis, a drag symbol appears. Simply click and drag to move along that axis.

Box Select

This icon (Figure 4) is only available for charts that include gauging points, or for detected rating shifts (less useful for that chart). When activated (Figure 5), it allows the user to draw a rectangle that selects the gauging points within it; the other points become more transparent. To move the rectangle, click *Pan*. The shape can also be resized using the white handles on its edges. To remove it entirely, refresh the page.



Figure 4: Box select icon selection

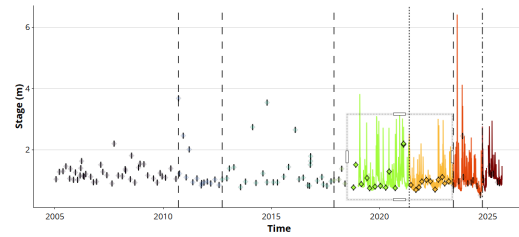


Figure 5: How to use the tool

Lasso Select

Similarly to *Box Select*, this tool (Figure 6) is only available for charts with gauging points, or for detected rating shifts. Unlike the previous tool, it allows the user to draw a free-form shape (Figure 7), which can be adjusted using the white handles. All other features of Box Select apply equally to Lasso Select.



Figure 6: Lasso select icon selection

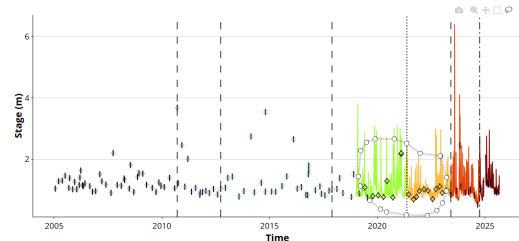


Figure 7: How to use the tool

Zoom In

There are several ways to zoom in on the graph (including using the icon shown in Figure 8), although using the mouse (by drawing a rectangle — Figure 9) is often more intuitive and quicker.

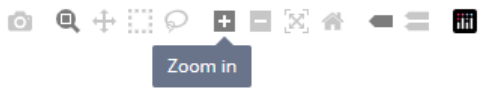


Figure 8: Zoom in on the graph

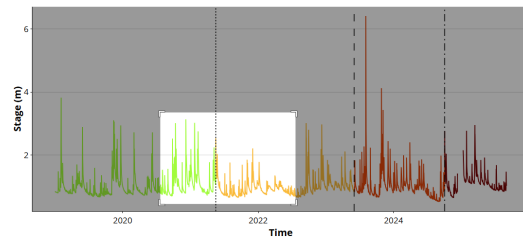


Figure 9: Zoom in using the mouse

It is also possible to use the mouse to zoom in on a specific part of the graph horizontally or vertically (see Figure 10 and Figure 11).

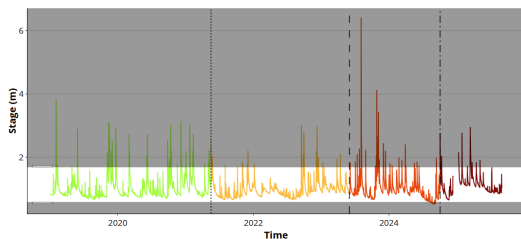


Figure 10: Zoom horizontally using the mouse

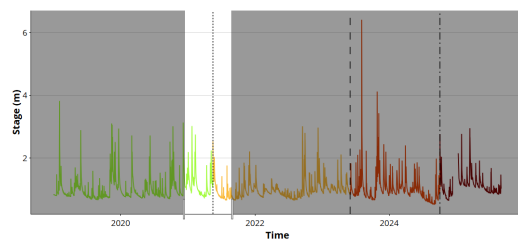


Figure 11: Zoom vertically using the mouse

Zoom Out

This icon (Figure 12) zooms out progressively. Double-clicking on the chart also returns to the initial view.

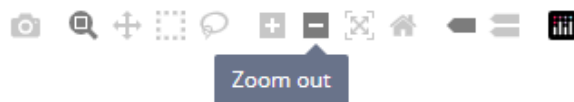


Figure 12: Zoom out on the graph

Autoscale

This option (Figure 13) lets the interface automatically select the most appropriate scale for displaying the chart.



Figure 13: Automatic scaling of plot

Reset Axis

This option (Figure 14) resets the chart to its original state, as displayed before any interaction.

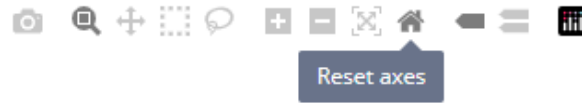


Figure 14: Reset of plot's axis

Show Closest Data on Hover

This tool (Figure 15) is used to inspect the data underlying the chart by hovering over it with the mouse. To view the labels of the rating shift detection methods, place the cursor at the top or bottom of the vertical axis. Otherwise, use the method described in the following section.

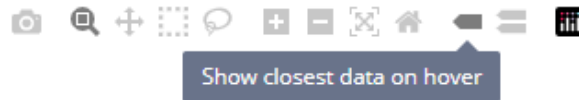


Figure 15: Inspecting data using the mouse

Compare Data on Hover

This icon (Figure 16) is used to compare the data underlying the chart by hovering over it with the mouse. Unlike the previous option, this one displays all labels simultaneously for nearby data points.

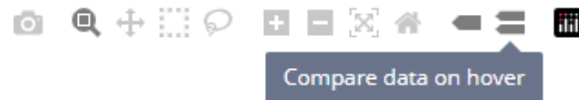


Figure 16: Comparing data using the mouse

Produced with plotly.js (v2.25.2)

This last icon (Figure 17) links to the website of the tool used to produce the charts.

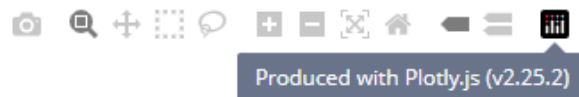


Figure 17: *Plotly* tool website

Contents

Station Location

The location of the station can be visualised on an interactive map.

Control Matrix

This matrix summarises the hydraulic analysis of the station, carried out using topographic maps, GIS (Geographic Information Systems — such as Google Earth, Géoportail, or QGIS®), cross-section surveys, and other data.

Controls correspond to the physical characteristics of the station that govern the water level at the studied cross-section. The type of control is specified in the matrix; for further details, refer to the BaRatinAGE website.

Segments represent portions of the watercourse governed by a constant number of controls.

Cells marked with "1" indicate the controls influencing the water level of each segment.

Parameters

The equation presented in the file is the generic equation used by BaRatinAGE:

$$Q = a(h - k)^c$$

Where: Q is the discharge, a depends on physical parameters, h is the water level, k is the activation height of the control, and c is a coefficient depending on the geometry of the control.

The table below summarises the parameters k , a , and c for each control, with a central value (median) and 95% credible intervals, allowing the software a margin of uncertainty around the central value. The last column indicates which parameter is variable (parameter c can never be variable). For k , it may be variable if the activation height of the control can be altered after a flood (for example, sediment deposition or erosion in the main channel will affect the activation height of that control). For a , it is considered variable if the width of the control, the slope, the roughness, etc., may be modified following a flood.

Detected Rating Shifts and Their 95% Uncertainties

The first lines in this graph show the shifts detected by each method, together with their 95% uncertainty (grey rectangles). Indeed, the shifts obtained do not always coincide with the flood peak or the exact date of calibration: the uncertainties are therefore necessary for the correct interpretation of the detected shifts. The various detection methods are:

1. **Gauging method:** uses the dispersion of gauging measurements to detect a rating shift.
2. **Recession method:** uses the stage records during periods of falling water levels, when the water level tends to align with the riverbed. When significant deviations are observed, a rating shift is identified.
3. **M6 or M7 modelling method:** this approach models the detected recessions before segmenting them.

The last row shows the shifts that were ultimately selected. It is possible that several shifts have been merged (to view the detection methods used for the selected calibration adjustments, hover the mouse over the vertical segments).

Dates of Selected Rating Shifts

This table lists all rating shifts detected by the BaM tool, along with the detection method used. Several methods are available, each relying on different data:

1. **Gauging method:** uses the scatter of gauging measurements to detect a rating shift.
2. **Recession method:** uses the limnograph during recession periods, during which the water level tends to align with the riverbed. When large discrepancies are observed, a rating shift is identified.
3. **M6 or M7 modelling method:** this approach models the detected recessions before segmenting them.

Limnograph by Detected Period

This chart displays the limnograph according to each detected period. Vertical markers are placed at the dates of the rating shifts, and the line style indicates the detection method used.

Hydrograph With Total Uncertainties

This chart displays the hydrograph computed by BaM for each period. Vertical markers are placed at the dates of the rating shifts, and the line style indicates the detection method. The grey shading represents the total uncertainties associated with the hydrograph computation.

Difference Between the BaM-Computed Hydrograph and the Provided Hydrograph (in Percentage)

This chart shows the differences between the two hydrographs under study — the one computed by BaM and the one computed by the hydrometric units (HU) — as a function of the periods detected by BaM. Vertical markers are placed at the dates of the rating shifts, and the line style indicates whether the observed shift was computed by BaM or by the station manager.

Percentage values may sometimes appear at the top or bottom of the chart when differences are too large to be shown at the current scale.

Absolute Difference Between the BaM-Computed Hydrograph and the Provided Hydrograph

This chart shows the absolute differences between the two hydrographs under study — the one computed by BaM and the one computed by the HU — as a function of the periods detected by BaM. Vertical markers are placed at the dates of the rating shifts, and the line style indicates whether the observed shift was computed by BaM or by the station manager.

Limnograph and Gaugings by Detected Period

This chart displays the limnograph and gauging measurements (with their stage uncertainties) according to each detected period. Vertical markers are placed at the dates of the rating shifts, and the line style indicates the detection method.

Hydrograph and Gaugings by Detected Period

This chart displays the hydrograph computed by BaM alongside the gauging measurements (with their discharge uncertainties), for each period. Vertical markers are placed at the dates of the rating shifts, and the line style indicates the detection method.

Gaugings

This chart displays the gauging measurements and their uncertainties on discharge according to the periods detected by BaM.

Rating Curves Provided and Computed with BaM

Two charts are presented in this section: the rating curves (RC) provided by the HU, and the rating curves computed with BaM. In both cases, gauging measurements (with their discharge uncertainties) are colour-coded according to the period to which they belong. For the BaM-computed rating curves, each curve is accompanied by an uncertainty band of the same colour as the corresponding period (colour overlaps are frequent).

Rating Curves Provided and Computed with BaM (Logarithmic Discharge Scale)

Two charts are presented in this section: the rating curves (RC) provided by the HU, and the rating curves computed with BaM, both with a logarithmic discharge scale. In both cases, gauging measurements (with their discharge uncertainties) are colour-coded according to the period to which they belong. For the BaM-computed rating curves, each curve is accompanied by an uncertainty band of the same colour as the corresponding period (colour overlaps are frequent).